

July 2021 Term
CSE 471 – Machine Learning
Class Test# 1 Time: 20 minutes

Name: _____ Student No. _____

No extra sheet shall be provided.

1. For the training set of the “WillWait” problem, calculate the information gain of the Type attribute.

Example	Input Attributes										Output
	Alt	Bar	Fri	Hun	Pat	Price	Rain	Res	Type	Est	WillWait
x ₁	Yes	No	No	Yes	Some	\$\$\$	No	Yes	French	0–10	y ₁ = Yes
x ₂	Yes	No	No	Yes	Full	\$	No	No	Thai	30–60	y ₂ = No
x ₃	No	Yes	No	No	Some	\$	No	No	Burger	0–10	y ₃ = Yes
x ₄	Yes	No	Yes	Yes	Full	\$	Yes	No	Thai	10–30	y ₄ = Yes
x ₅	Yes	No	Yes	No	Full	\$\$\$	No	Yes	French	>60	y ₅ = No
x ₆	No	Yes	No	Yes	Some	\$\$	Yes	Yes	Italian	0–10	y ₆ = Yes
x ₇	No	Yes	No	No	None	\$	Yes	No	Burger	0–10	y ₇ = No
x ₈	No	No	No	Yes	Some	\$\$	Yes	Yes	Thai	0–10	y ₈ = Yes
x ₉	No	Yes	Yes	No	Full	\$	Yes	No	Burger	>60	y ₉ = No
x ₁₀	Yes	Yes	Yes	Yes	Full	\$\$\$	No	Yes	Italian	10–30	y ₁₀ = No
x ₁₁	No	No	No	No	None	\$	No	No	Thai	0–10	y ₁₁ = No
x ₁₂	Yes	Yes	Yes	Yes	Full	\$	No	No	Burger	30–60	y ₁₂ = Yes

2. Calculate sensitivity and specificity based on the below confusion matrix.

		Predicted Class	
		Churn=True	Churn=False
Actual Class	Churn = True	15	5
	Churn = False	9	22